الرواد الرقميون
DiGiLiANS

وزارة الاتصالات وتكنولوجيا المعلومات

Enterprise Banking
Risk Management
Analytics Solution

Credit Risk Analytics, Default Prediction & Expected Credit Loss

Start



project date 9 September 2026

Enterprise Banking
Risk Management
Analytics Solution

Credit Risk Analytics, Default Prediction & Expected Credit Loss

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September 2026

Presentation Storyline

01 WHY & SCOPE
Nesma — Business problem • stakeholders • project boundaries

02 DATA FOUNDATION
Maysoun — Sources • profiling • Bronze & Silver

03 ANALYTICAL MODEL
Aya — Gold views • relationships • Dashboard 1

04 CURRENT RISK
Omnia — Dashboards 2–4 • delinquency • early warning • collections

05 PREDICTIVE RISK
Nedaa — PD models • evaluation • segmentation • Dashboard 5

06 DECISION & ACTION
Nesma — XAI • LGD/EAD/ECL • stress testing • watchlist

WHY → DATA → STRUCTURE → INSIGHT → PREDICTION → ACTION

Team & Presentation Flow

Flow: Nesma → Maysoun → Aya → Omnia → Nedaa → Nesma



Nesma Saleh Abd Elgwad Mohamed

Story & Risk Decisions

Intro, stakeholder value, final risk layer & closing.



Maysoun Adel Mahmoud Ahmed

Data Foundation

Sources, profiling, Bronze & Silver.



Aya Mamdouh Ebrahim Abdelkader

Gold & Portfolio

Gold views, model, relationships & Dashboard 1.



Omnia Mamdouh Bekhet Abo Haroun

Risk Dashboards

Dashboards 2–4: delinquency, early warning & collections.



Nedaa Mohamed Abd El Basset

Predictive Risk

ML modeling, evaluation, segmentation & Dashboard 5.

Supervision: Prof. Dr. Rania El Gohary
Track Head: Dr. Ahmed Zakaria
Academic Team: Dr. Yasser Ali
Instructor: Dr. Wael Mahrous (NXT)

Business Problem & Decision Need

Presenter: Nesma | Why are we doing this?

THE BUSINESS CHALLENGE

- Risk data spans loans, customers, payments, collections and risk events.
- Fragmented reporting can create inconsistent KPI definitions and slow analysis.
- Management needs one governed view that links exposure, deterioration, prediction and action.

QUESTIONS THE SOLUTION MUST ANSWER

- Where is the bank's money concentrated?
- Where are delinquency and default increasing?
- What is driving risk and weak collection performance?
- Which loans may become risky next?
- What should the bank do first?

Business Need: One reliable analytical view of portfolio performance and credit risk.

Objectives, Stakeholders & Scope

Presenter: Nesma | What does success look like?

OBJECTIVES

- Standardize 13 banking datasets in governed SQL layers.
- Publish six Gold views: three dimensions and three facts.
- Build five connected Power BI dashboards.
- Compare Logistic Regression and Decision Tree; segment with K-Means.
- Connect analysis to risk actions and expected-loss concepts.

STAKEHOLDERS

- Senior Management
- Risk Management
- Credit Department
- Collections
- Branch Management
- Finance / Provisioning

SCOPE GUARDRAIL

- In scope: 30,000 loans, 22,000 customers, portfolio/risk/collections analytics and an ML prototype.
- Analytical ECL framework is included for decision support.
- Out of scope: confidential production data, live deployment and a fully governed regulatory IFRS 9 engine.

Data Sources & Business Requirements

Presenter: Maysoun | Can we trust the data we are about to analyze?

DATA DOMAINS

- Customers and credit profile
- Loans and loan products
- Branches and loan officers
- Payments and cashflow
- Delinquency, defaults and risk events
- Model-ready features for predictive analysis

Prototype data is simulated/controlled for analytics and does not represent confidential bank records.

BUSINESS QUESTION → REQUIRED DATA

- Portfolio → amounts, outstanding balance, product and branch.
- Delinquency / Default → DPD, overdue status, default flag and default amount.
- Collections → due amount, paid amount, late status and payment behavior.
- Prediction → only features available before the outcome to avoid leakage.

Data Profiling, Bronze & Silver

Presenter: Maysoun | From raw data to trusted data

1. PROFILE

Understand the data before changing it

- Nulls & true duplicates
- Data types & ranges
- Key / date consistency
- Referential integrity

2. BRONZE

Preserve the raw source

- Raw landing layer
- Minimum transformation
- Traceability to source
- Recovery / audit reference

3. SILVER

Clean and standardize

- Correct types and dates
- Standardize IDs and flags
- Validate business fields
- Prepare trusted data for Gold

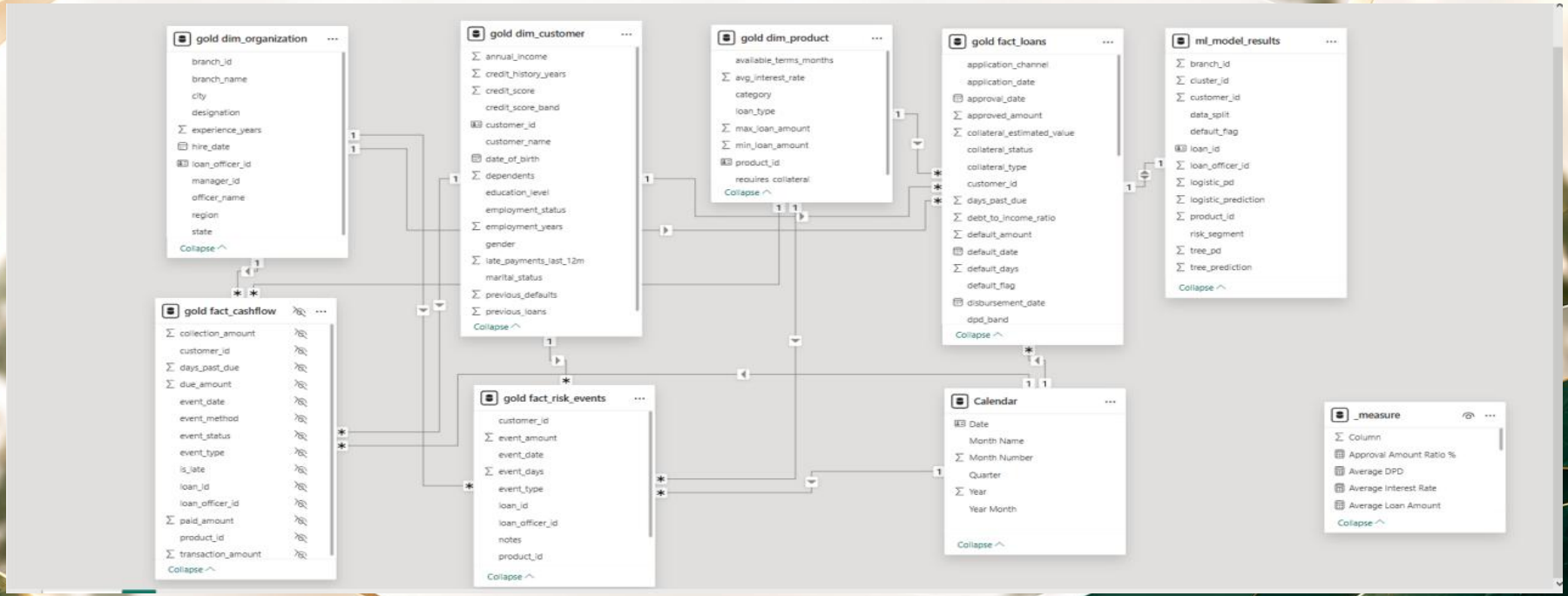
Validation checks: row counts • unique keys • amount totals • valid dates • consistent categories

Output: standardized Silver data ready for the business-ready analytical layer.

Gold Layer & Analytical Data Model

Presenter: Aya | How did we structure the data?

Six Gold views: 3 dimensions + 3 facts | Shared Calendar | Loan-level ml_model_results



Dimensions (1) filter Facts (*) • Different grains prevent duplication • Centralized measures keep KPIs consistent

Dashboard 1 — Portfolio Overview & Concentration

Presenter: Aya | How large is the portfolio, and where is the money concentrated?

30K

Total Loans

Distinct loan portfolio

26.86B EGP

Total Loan Amount

Portfolio size

24.44B EGP

Approved Amount

Approved exposure

12.75B EGP

Outstanding Balance

Remaining exposure

Concentration

Product Share

Identify concentration risk

Growth + Pricing

Monthly Trend

Loan amount vs avg. interest rate

Portfolio Overview & Concentration

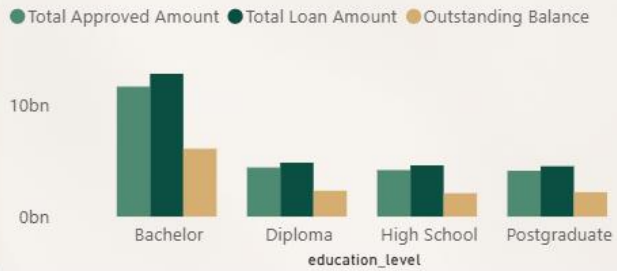
loan_type: All

region: All

- Overview
- Portfolio Overview & Concentration
- Delinquency & Default Risk
- Risk Drivers & Early Warning
- Collections & Recovery
- Predictive Risk & Customer Segmentation

Total Loans	Total Loan Amount	Total Approved Amount	Outstanding Balance
30K	27bn	24bn	13bn

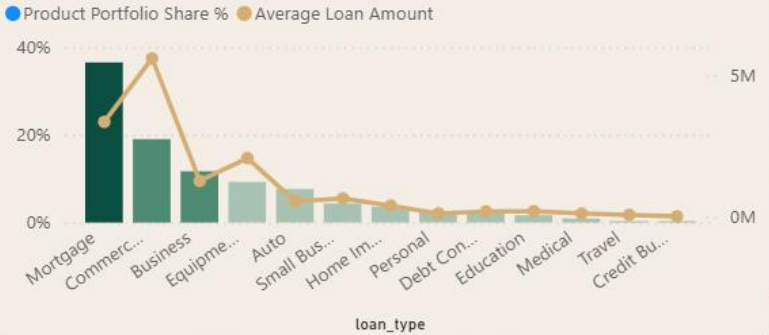
Portfolio Financial Position



Branch Exposure Analysis



Product Concentration by Loan Type



Loan Growth & Pricing Trend



Dashboard 2 — Delinquency & Default Risk

Presenter: Omnia | How much of the portfolio is showing repayment problems?

DELINQUENCY VIEW

20.8%	6K	112.55 M EGP
Delinquency Rate	Overdue Loans	Overdue Amount

- DPD bands: 1–30 • 31–60 • 61–90 • 90+ days
- Average DPD is evaluated on loans with DPD > 0.

DEFAULT VIEW

8.57%	2,572	1.39B EGP
Default Rate	Defaulted Loans	Default Amount

- Track default trend over time and compare loan types.
- Branch risk combines default rate with outstanding exposure.

Decision lens: combine risk severity with financial exposure—never rank risk using one metric alone.

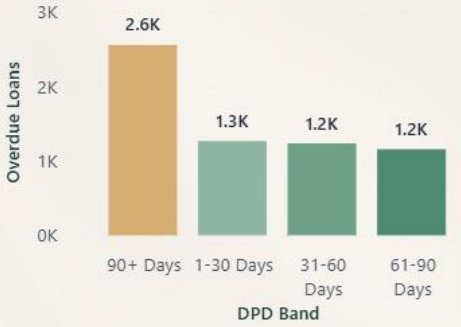
Delinquency & Default Risk

Year: All | loan_type: All | branch_name: All

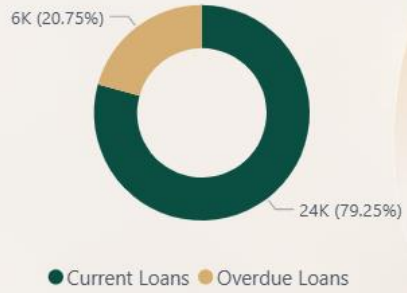
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Total Overdue Amount	Overdue Loans	Delinquency Rate %	Defaulted Loans
112.55M	6K	21%	3K

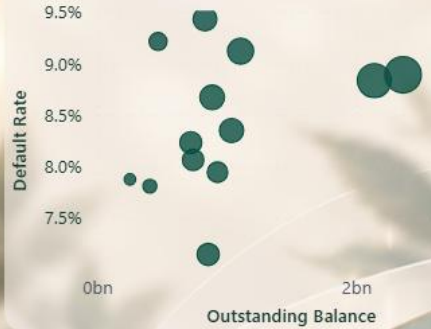
Overdue Loans by DPD Band



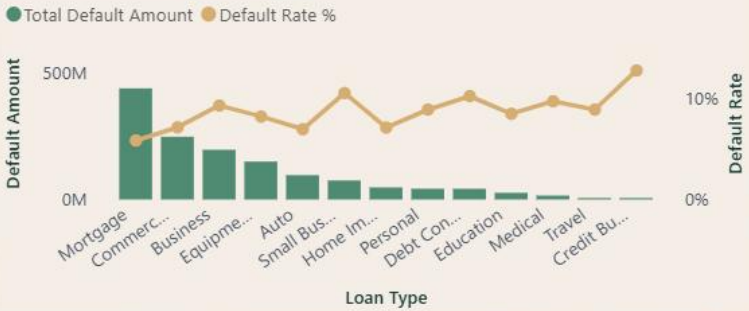
Current vs Overdue Loans



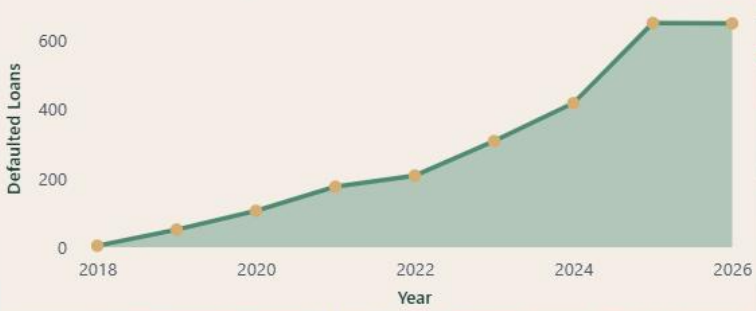
Branch Risk Exposure



Default Risk by Loan Type



Defaulted Loans by Year



Dashboard 3 — Risk Drivers & Early Warning

Presenter: Omnia | Where is the risk coming from, and can we detect deterioration earlier?

RISK DRIVERS

- Credit Score Band
- Loan Type & Product
- Branch / organization
- DPD and payment behavior
- Previous risk history

EARLY WARNING

- Detect deterioration before default
- Repeated late-payment signals
- High / rising DPD cases
- Prioritize cases for review
- Support proactive intervention

IFRS 9 STAGING LENS

- Stage 1: performing exposures
- Stage 2: significant increase in credit risk
- Stage 3: credit-impaired exposures
- DPD is an indicator / backstop
- Staging is not determined by DPD alone

Purpose: move from 'Where is the risk?' to 'Why is it happening, and who needs attention early?'

Risk Drivers & Early Warning

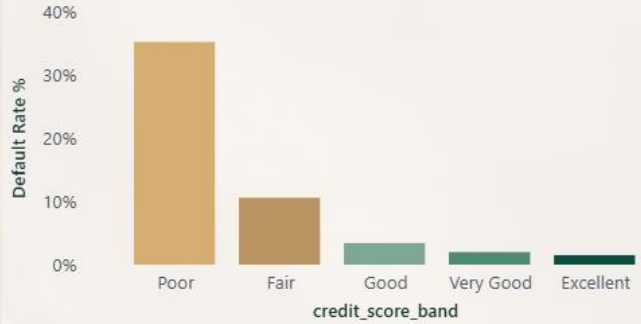
Year: All

loan_type: All

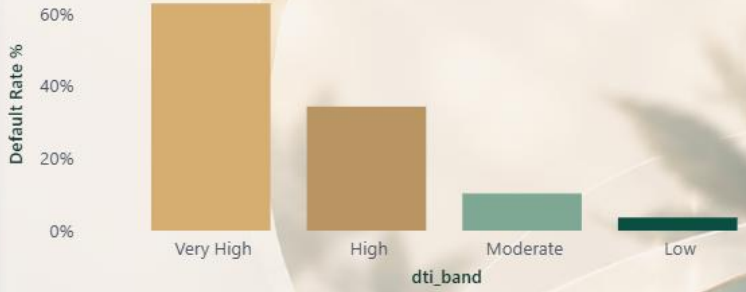
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Avg Score of Def.	Avg DTI of Defaulters	Repeat Risk Rate %	Default Conversion Rate %
605	28.2%	17%	41.3%

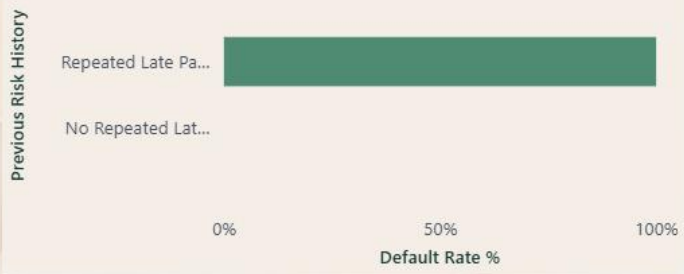
Credit Score Band vs Default Rate



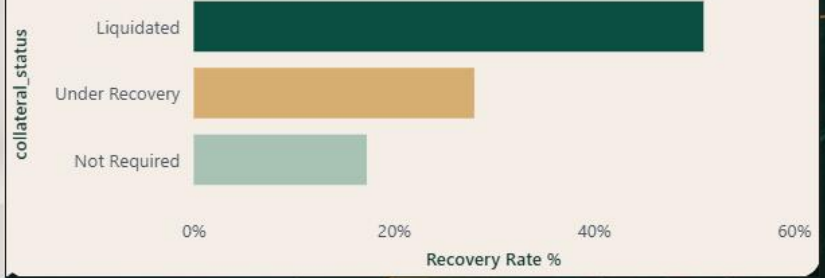
DTI Band vs Default Rate



Default Rate by Previous Risk History



Recovery Rate by Collateral Status



Dashboard 4 — Collections & Recovery

Presenter: Omnia | Is the bank collecting what is due, and which cases need priority?

459.28M EGP

Total Due Amount
Scheduled / due cashflow

454.02M EGP

Total Paid Amount
Collected cashflow

98.9%

Collection Rate
Paid ÷ Due

ANALYSIS

- Due vs paid trend over time
- Collection gap by product / branch
- Late-payment frequency vs default risk

ACTION — URGENT COLLECTION CASES

- Prioritize by DPD + unpaid amount + exposure
- Use repayment behavior to focus collection resources

A high overall collection rate can still hide risky pockets—drill down before concluding performance is strong.

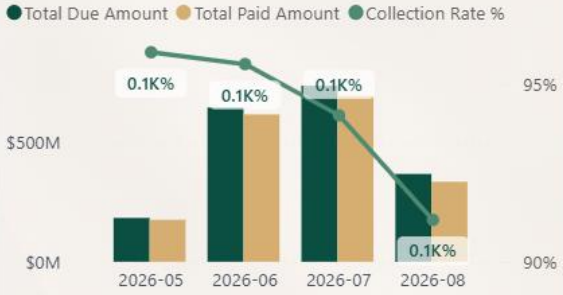
Collections & Recovery

Year: All | loan_type: All | branch_name: All

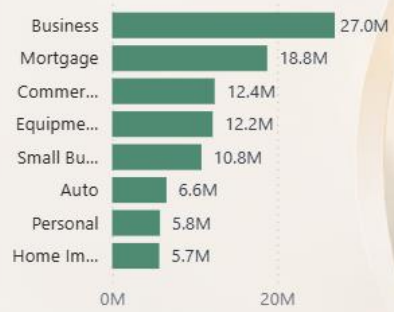
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Total Due Amount	Total Paid Amount	Collection Rate %	Collection Gap Amount
$1.9bn	$1.83bn	94.2%	112.5M

Monthly Collection Performance



Collection Gap by Loan Product



Collection Rate by Branch



Payment Performance by Method



Urgent Collection Cases

Loan ID	Loan Type	Due	Total Paid Amount	Gap Amount	Max of days_past_due
1000001	Auto	$27,790.7	$16,966.03	10,824.7	125
1000040	Auto	$41,892.2	$20,296.46	21,595.7	160
1000337	Auto	$21,543.6	$19,671.66	1,871.9	27
1000359	Auto	$6,527.8	$5,484.62	1,043.2	27
1000428	Auto	$64,851.2	$0.00	64,851.2	115
1000434	Auto	$24,373.1	$18,347.65	6,025.4	13
1000455	Auto	$40,974.9	$4,981.78	35,993.1	83
1000458	Auto	$28,773.4	$15,003.11	13,770.3	78
Total		$245,108,288.1	$132,562,734.72	112,545,553.4	327

Machine Learning — Prediction & Evaluation

Presenter: Nedaa | What may happen next?

0 / 1

Target: Default Flag

Binary classification

LR + DT

Models Compared

Logistic Regression • Decision Tree

82.35%

Logistic ROC-AUC

Discrimination performance

73%

Default Recall

Default cases identified

Pre-outcome Features → **Train / Test Split** → **Model Comparison** → **Probability of Default (PD)**

Evaluate with Precision • Recall • F1 • Confusion Matrix • ROC-AUC — Accuracy alone is not enough

Leakage control: use only information available before the default outcome.

Dashboard 5 — Predictive Risk & Customer Segmentation

Presenter: Nedaa | Turn model outputs into a monitoring view

WHAT THE DASHBOARD BRINGS TOGETHER

- Loan-level model outputs are exported to ml_model_results.
- Predicted PD ranks loans by estimated default likelihood.
- K-Means provides clusters; after profiling, business labels are Low / Medium / High Risk.
- The view combines PD, risk segment, exposure and actual status for proactive monitoring.

MODEL-LED SIGNALS

4,202

High-Risk Loans

31.65%

High-Risk Default Rate

PD is a probability—not certainty. The model supports credit and risk judgment; it does not replace it.

Predictive Risk & Customer Segmentation

loan_type: All

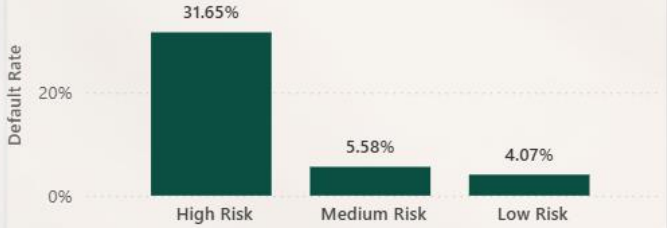
branch_name: All

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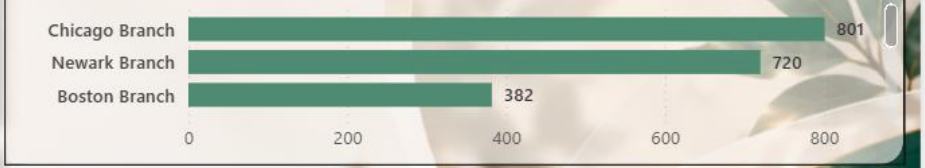
High Risk Loans	High Risk Share %	High Risk Def Rate %	Watchlist Loans	High-Risk Exposure
4K	14%	31.7%	3K	1.75bn

LGD %	Net Credit Loss	Average PD
66%	917M	37.6%

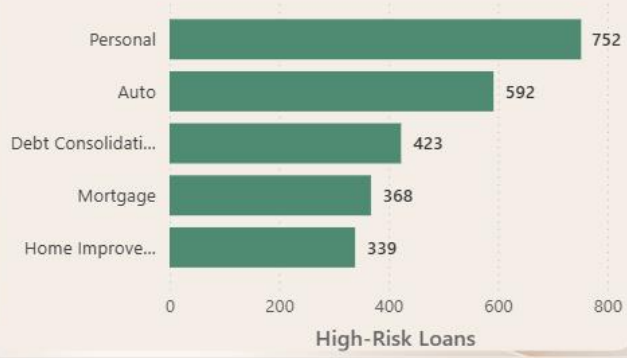
Default Rate by Risk Segment



High-Risk Loans by Branch



Top 5 Products by High-Risk Loans



Early-Warning Watchlist

Loan ID	Customer ID	Loan Type	Branch	credit_score	DTI	DPD	Outstanding Balance	Risk_segr
1025540	5755	Education	Newark Branch	483	0.57	0	188700	High Risk
1024296	4910	Debt Consolidation	Miami Branch	482	0.56	46	155500	High Risk
1026954	16262	Personal	New York Branch	542	0.53	0	0	High Risk
1013173	12194	Auto	New York Branch	514	0.53	20	402300	High Risk
1025426	21710	Personal	Newark Branch	500	0.52	0	128900	High Risk

Final Risk Layer — From Prediction to Action

Presenter: Nesma | What should the bank do?

Analytical framework: quantify potential loss, test adverse scenarios and prioritize cases for action.

EXPLAINABLE AI

Why is this exposure risky?
Explain prediction drivers.

LGD

Loss Given Default
How much may be lost if default occurs?

EAD

Exposure at Default
What exposure is expected at default?

ECL

Expected Credit Loss
Conceptually combines PD × LGD × EAD.

STRESS TESTING

How do losses change under adverse assumptions?

FINAL WATCHLIST

Prioritized cases + risk drivers + recommended actions.

Full governed IFRS 9 production implementation requires additional scenarios, discounting, validation and model governance.

Business Value, Actions & Limitations

Presenter: Nesma | Close the loop from insight to decision

BUSINESS VALUE

- One governed view across portfolio, risk and collections.
- Earlier identification of deteriorating exposures.
- Focused collection and credit-review priorities.
- Consistent KPI definitions from SQL to Power BI.

RISK ACTIONS

- Low Risk → normal monitoring.
- Medium Risk → enhanced monitoring and early contact.
- High Risk → priority review, collection action and exposure / collateral review.
- Use model outputs as decision support—not automatic approval or rejection.

LIMITATIONS / NEXT STEP

- Prototype uses simulated / controlled data.
- Production use requires security, validation and live integration.
- Add macroeconomic scenarios, drift monitoring and formal IFRS 9 governance.
- Maintain an auditable KPI and model catalogue.

FINAL MESSAGE: Fragmented Data → Governed Insight → Prediction → Prioritized Risk Action

Benchmark Reference – Egyptian Banking Sector

Regulatory Framework & Key Indicators

We use official Egyptian references from the Central Bank of Egypt (CBE) and IFRS 9 guidelines to provide context for our risk analysis.

البنك المركزي المصري
Central Bank of Egypt
STABILITY | TRUST | SUSTAINABLE GROWTH



Default Definition (90 Days)

According to IFRS 9 and CBE Guidelines

90 DAYS

A borrower is considered in default (Stage 3 / credit-impaired) when payments are 90 days past due.

- The 90 days backstop is consistent with IFRS 9.
- Applied in the Egyptian banking sector under CBE instructions.
- Qualitative indicators can also be used to identify default earlier (e.g., financial distress, restructured loans, unlikeliness to pay).

For SMEs, the temporary exception was phased out, and from January 2026, the 90-day criteria fully applies.

Key Egyptian Banking Benchmark

Central Bank of Egypt – Latest Available Data

Non-Performing Loans (NPL) Ratio

1.9% of total loans

End of Q4 2025

Provision Coverage Ratio

90.2% (Total provisions / NPLs)

End of Q4 2025

Source

Central Bank of Egypt (CBE)
Financial Soundness Indicators
Press Release – March 2026
www.cbe.org.eg

Note: These figures represent the overall Egyptian banking sector and are used as an external reference, not a direct comparison with our project KPIs unless definitions are aligned.

How We Use the Benchmark

In Our Project

1. Use 90+ DPD as the default backstop in our analysis (aligned with IFRS 9 & CBE).
2. Compare key risk indicators with the sector NPL ratio (1.9%) as an external reference, not a direct comparison.
3. Ensure KPI definitions and denominators are consistent before making like-for-like comparisons.
4. Use the benchmark to provide context, set risk appetite, and support recommendations.

Important

Our project Default Rate (8.57%) is calculated as number of defaulted loans + total loans. The CBE NPL ratio (1.9%) is usually based on exposure amounts and may include different classifications, so direct comparison is not appropriate without harmonizing the definitions.

Purpose | To use trusted regulatory benchmarks to validate our approach, provide context for our findings, and support data-driven recommendations.

— From Data to Safer Decisions

References & Resources

Academic & Technical References

- [1] Microsoft. Power BI documentation: modeling, DAX, report design and refresh. Microsoft Learn.
- [2] Microsoft. SQL Server documentation: relational design, views, data types and query validation. Microsoft Learn.
- [3] IFRS Foundation. IFRS 9 Financial Instruments.
- [4] Basel Committee on Banking Supervision. Basel Framework. Bank for International Settlements.

Tools, Datasets & Repositories

- Power BI: https://learn.microsoft.com/power-bi/
- SQL Server: https://learn.microsoft.com/sql/sql-server/
- IFRS 9: https://www.ifrs.org/issued-standards/list-of-standards/ifrs-9-financial-instruments/
- Basel Framework: https://www.bis.org/basel_framework/
- GitHub: https://github.com/nedaaalgdawy-svg

Project evidence includes SQL scripts, validation queries, the Power BI semantic model, KPI definitions, ML outputs and implementation screenshots.

Thank You!

Team Name: Risk Wise

Track: Professional Data Analysis

GitHub: https://github.com/nedaaalgdawy-svg

From Data → Insight → Prediction → Action